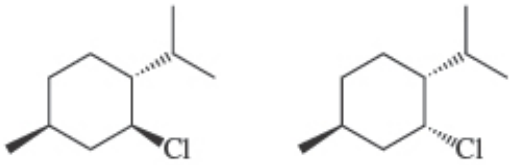
Organic Cumulative Exam September 8, 2023

All problems were taken from Organic Chemistry by Carey & Giuliano, 9th edition.

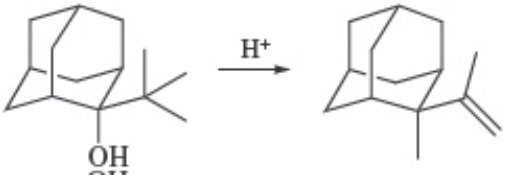
5.43 Menthyl chloride and neomenthyl chloride have the structures shown. One of these stereoisomers undergoes elimination on treatment with sodium ethoxide in ethanol much more readily than the other. Which reacts faster, menthyl chloride or neomenthyl chloride? Why?



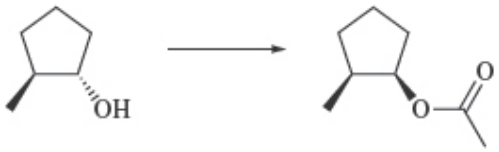
Menthyl chloride Neomenthyl chloride

5.48 Write a sequence of steps depicting the mechanisms of each of the following reactions. Use curved arrows to show electron flow.

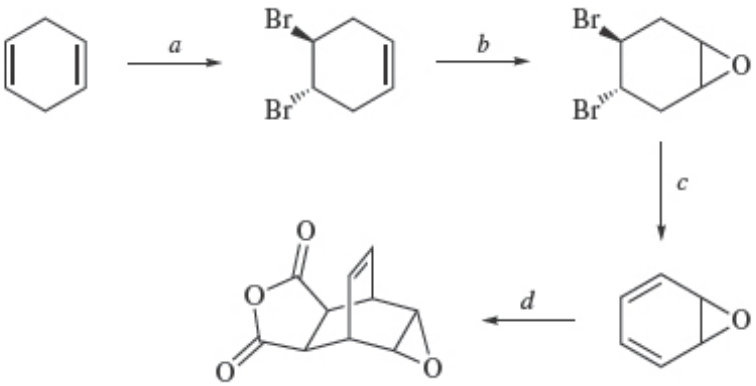
a.



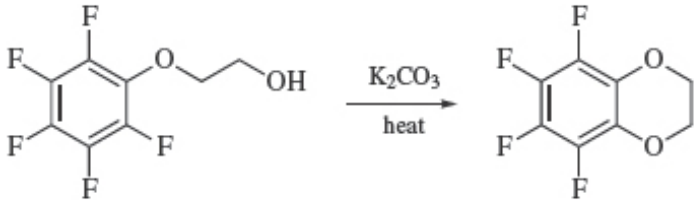
8.39 a. Suggest a reasonable series of synthetic transformations for converting *trans*-2-methylcyclopentanol to *cis*-2-methylcyclopentyl acetate.



10.47 Suggest reagents suitable for carrying out each step in the following synthetic sequence:

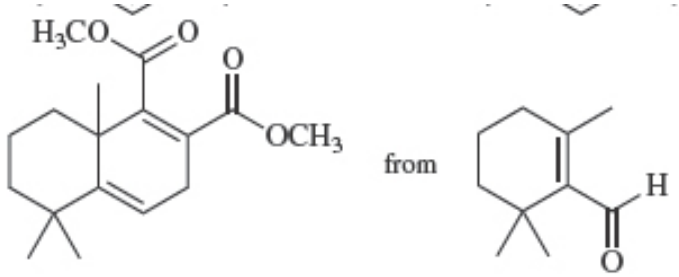


12.57 Suggest a reasonable mechanism for the following reaction:

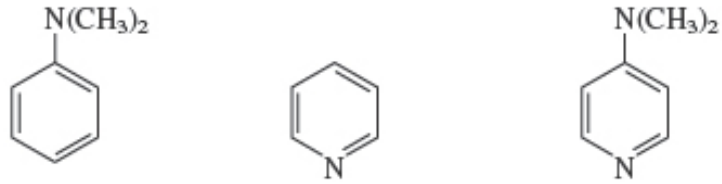


17.44 Syntheses of each of the following compounds have been reported in the chemical literature. Using the indicated starting material and any necessary organic or inorganic reagents, describe short sequences of reactions that would be appropriate for each transformation.

g.



21.53 *N,N*-Dimethylaniline and pyridine are similar in basicity, whereas 4-(*N,N*-dimethylamino) pyridine is considerably more basic than either.



N,N-Dimethylaniline
pK_a of conjugate acid = 5.1

Pyridine
pK_a of conjugate acid = 5.3

4-(*N,N*-Dimethylamino)pyridine
pK_a of conjugate acid = 9.7

Apply resonance principles to identify the more basic of the two nitrogens of 4-(*N,N*-dimethylamino)pyridine, and suggest an explanation for its enhanced basicity.